

Module 08: Planning in Crochet Circles

Planning a Color Palette

Learning Objectives

1. Describe color palette design as an expected free energy minimization process.
2. Explain dye lot management and project-yarn matching as planning under uncertainty.
3. Connect yardage estimation and material planning to the planning component of Active Inference.

Introduction

Before the first stitch is made, the crocheter plans. They choose a pattern, select a yarn, design a color palette, estimate yardage, and imagine the finished piece. This planning process — imagining futures, evaluating options, managing uncertainty — is the capstone of the Active Inference framework. It is where everything comes together: the systems knowledge, the material expertise, the perceptual skill, the cognitive models, the action capabilities, the learning history, and the communicative understanding.

In Active Inference, **planning** is the process of evaluating possible future actions by simulating their expected outcomes and choosing the action sequence that minimizes **expected free energy** — the combination of achieving goals and reducing uncertainty. This module explores how crocheters plan projects through the lens of Active Inference.

Key Concepts

1. Color Palette Design as Expected Free Energy Minimization

Choosing colors for a project is one of the most creatively demanding planning tasks in crochet. The crocheter must imagine how multiple colors will interact in a fabric they have not yet made.

In Active Inference terms, color palette design is **expected free energy minimization**. The crocheter mentally simulates different color combinations and evaluates each one: - **Will these colors create the emotional tone I want?** (warm, cool, vibrant, subtle) - **Will they have enough contrast to show the stitch pattern?** (too-similar colors will obscure texture) - **Will the viewer's eye move across the piece as intended?** (color placement guides visual flow) - **Will I still like this palette after hundreds of hours of crocheting?** (predicting future preferences)

The crocheter chooses the palette that minimizes expected surprise — the one where the finished piece is most likely to match their vision.

2. Dye Lot Management

Dye lot management is planning under real uncertainty. The crocheter must buy enough yarn of the same dye lot to complete their project, because different dye lots may not match.

This requires: - **Estimating total yardage** needed (a prediction about the future) - **Adding a safety margin** (acknowledging uncertainty in the estimate) - **Buying all at once** (reducing the risk of dye lot unavailability) - **Planning for contingencies** (what if the yarn is discontinued?)

Dye lot management is a microcosm of planning under uncertainty — the crocheter cannot know exactly how much yarn they will need, so they must plan with buffers and contingencies.

3. Project-Yarn Matching

Not every yarn is suitable for every project. Matching a yarn to a project requires the crocheter to plan across multiple dimensions: - **Fiber suitability**: Is this the right fiber for the intended use? (Wool for warmth, cotton for kitchen items, acrylic for baby items that need frequent washing) - **Weight appropriateness**: Is this yarn weight compatible with the pattern's gauge? - **Texture compatibility**: Will this yarn's texture complement or fight the stitch pattern? - **Care requirements**: Will the finished item's care needs match the recipient's lifestyle? - **Budget planning**: Can I afford enough of this yarn for the entire project?

Each of these evaluations is a prediction about future outcomes. The crocheter who plans well minimizes the chance of an unpleasant surprise in the middle of or after completing the project.

4. Yardage Estimation

Estimating how much yarn a project will require is a planning challenge that combines material knowledge with mathematical prediction: - **Pattern yardage** provides a starting estimate (but assumes a specific gauge) - **Gauge differences** require adjustment (tighter gauge uses more yarn per area) - **Stitch pattern** affects consumption (open stitches use less yarn; tall stitches use more) - **Size modifications** add or subtract yardage - **Safety margin** accounts for estimation uncertainty (typically 10-20% extra)

Experienced crocheters become skilled yardage estimators because their generative models include predictions about yarn consumption rates for different stitch types and fiber properties. This is planning with a well-trained model.

5. The Iterative Planning Cycle

Crochet planning is not a one-shot process. It is iterative: 1. **Initial plan**: Choose pattern, yarn, colors, estimate yardage 2. **Swatch**: Test the plan with a small sample 3. **Revise**: Update the plan based on swatch results (change hook size, adjust color choices, recalculate yardage) 4. **Begin**: Start the project with the revised plan 5. **Monitor**: Check progress against the plan as work proceeds 6. **Adapt**: Make adjustments as needed (change colors, adjust sizing, manage remaining yardage)

This iterative cycle is Active Inference in action: plan, act, observe, revise. The crocheter's plan is a generative model of the future project, and each step of execution provides evidence that updates the plan.

The Free Energy Principle and Planning

The Free Energy Principle tells us that planning is the process of simulating future action sequences and choosing the one that minimizes expected free energy — the combination of goal achievement (pragmatic value) and information gain (epistemic value).

In crochet planning: - **Pragmatic value:** Will this plan produce the object I want? Will it be beautiful, functional, and well-crafted? - **Epistemic value:** Will this plan teach me something new? Will it reduce my uncertainty about a new technique or material?

The best projects often balance both — they are achievable enough to succeed (low pragmatic surprise) and challenging enough to grow the crocheter's skills (high epistemic value).

Conclusion

Planning a crochet project — from color palette to yardage estimation to dye lot management — is Active Inference at its most comprehensive. The crocheter draws on everything they have learned about systems, agents, perception, cognition, action, learning, and communication to imagine a future object and chart a path toward making it real. This is the creative power of the fiber arts: the ability to hold a vision in mind, work through uncertainty with skill and material knowledge, and bring something new into the world, one stitch at a time.

Connections

- **Previous Module:** [Communication — The Language of Fiber and Color](#)
- **Course Home:** [Fiber & Flow](#)